4.5.3 The inverse tangent function and its properties

The tangent function is not one-to-one in the entire domain $\mathbb{R} \setminus \left\{\frac{\pi}{2} + k\pi, k \in \mathbb{Z}\right\}$. However, $\tan x : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to \mathbb{R}$ is a bijective function. Now, we define the inverse tangent function with $\mathbb{R}$ as its domain and $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ as its range.

Definition 4.5

For any real number x, define $\tan^{-1} x$ as the unique number y in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ such that $\tan y = x$. In other words, the inverse tangent function $\tan^{-1} : (-\infty, \infty) \to \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ is defined by $\tan^{-1}(x) = y$ if and only if $\tan y = x$ and $y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

From the definition of $y = \tan^{-1} x$, we observe the following:

(i) $y = \tan^{-1} x$ if and only if $x = \tan y$ for $x \in \mathbb{R}$ and $-\frac{\pi}{2} < y < \frac{\pi}{2}$.

(ii) $\tan\left(\tan^{-1} x\right) = x$ for any real number x and $y = \tan^{-1} x$ is an odd function.

(iii) $\tan^{-1}\left(\tan x\right) = x$ if and only if $-\frac{\pi}{2} < x < \frac{\pi}{2}$. Note that $\tan^{-1}\left(\tan \pi\right) = 0$ and not π.

Note

(i) Whenever we talk about inverse tangent function, we have,
$\tan : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to \mathbb{R}$ and $\tan^{-1} : \mathbb{R} \to \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

(ii) The restricted domain $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ is called the **principal domain** of tangent function and the values of $y = \tan^{-1} x$, $x \in \mathbb{R}$, are known as **principal values** of the function $y = \tan^{-1} x$.

4.5.4 Graph of the inverse tangent function

$y = \tan^{-1} x$ is a function with the entire real line $(-\infty, \infty)$ as its domain and whose range is $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Note that the tangent function is undefined at $-\frac{\pi}{2}$ and at $\frac{\pi}{2}$. So, the graph of $y = \tan^{-1} x$ lies strictly between the two lines $y = -\frac{\pi}{2}$ and $y = \frac{\pi}{2}$, and never touches these two lines. In other words, the two lines $y = -\frac{\pi}{2}$ and $y = \frac{\pi}{2}$ are horizontal asymptotes to $y = \tan^{-1} x$.

Fig. 4.17 and Fig. 4.18 show the graphs of $y = \tan x$ in the domain $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and $y = \tan^{-1} x$ in the domain $(-\infty, \infty)$, respectively.

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